

UX-0004 — Butlerly Core User Flows

Version 1.0

Purpose

This document defines the detailed Version 1 user flows for Butlerly. It converts the journeys in UX-0002 and the structure in UX-0003 into step-by-step interaction sequences that can guide wireframes, prototypes, engineering implementation, analytics, accessibility review, and usability testing.

The flows are designed for one product with three native experiences: smartphone, tablet, and desktop. The same goals, records, permissions, and system states apply across devices, while layouts and interaction density adapt to each platform.

1. Flow Design Rules

All Version 1 flows must follow these rules:

- Start from a clear user intent.
- Minimize decisions before the user receives value.
- Use AI to propose before asking the user to fill a blank form.
- Ask one focused question when clarification is required.
- Keep local functionality available when the network or an external provider is unavailable.
- Explain when data will leave the device.
- Preserve the original source when extraction or interpretation fails.
- Make reversible actions easy to undo.
- Merge non-competing cross-device changes automatically.
- Use timestamps only when the same field of the same record changes on different devices.
- Preserve meaningful version history for recovery.
- Present one clear primary action at each decision point.
- Provide keyboard, screen-reader, touch, and pointer equivalents where applicable.

2. Flow Notation

The following notation is used:

- [Screen or State] — a visible user interface state.
- (User Action) — an explicit user action.
- {System Action} — a Butlerly background or automated action.
- <Decision> — a branching condition.
- — normal progression.
- ↺ — retry or recovery.
- ✓ — successful completion.
- ! — exception or attention state.

3. Flow Overview

The required Version 1 flows are:

- F01 — First Launch and Trust Establishment
- F02 — Capture a Receipt on Smartphone
- F03 — Import a Digital Receipt through Share Sheet
- F04 — Import Files or Statements
- F05 — AI Processing and Automatic Completion
- F06 — Review a Low-Confidence Item
- F07 — Resolve a Possible Duplicate
- F08 — Search and Retrieve a Record
- F09 — Ask Butlerly About a Record
- F10 — Edit, Undo, and Restore
- F11 — Add a Trusted Device
- F12 — Synchronize and Merge Changes
- F13 — Revoke a Lost Device
- F14 — Use Butlerly Offline
- F15 — Manage Privacy and External AI Consent
- F16 — Export Data
- F17 — Delete and Restore Data
- F18 — Recover from Processing, Import, or Sync Failure

4. F01 — First Launch and Trust Establishment

Goal

Begin using Butlerly while understanding local storage, optional synchronization, and external processing.

Primary entry

First application launch.

Flow

[Welcome]

→ Butlerly states the core promise in one sentence.

→ (Continue)

→ [Privacy Summary]

→ Butlerly explains:

- Data is stored locally by default.
- Synchronization is optional.
- External AI requires explicit permission.
- Export and deletion remain under user control.

- (Continue)
- [Choose a First Action]
 - Capture a receipt.
 - Import a file.
 - Explore sample data.
- <Selected action requires permission?>
 - Yes → [Contextual Permission Explanation]
 - (Allow or Not Now)
 - <Permission granted?>
 - Yes → Continue selected action.
 - No → Offer non-blocking alternative.
 - No → Continue selected action.
- User completes one useful task.
- [Completion and Optional Setup]
- Offer, but do not require:
 - Enable Apple synchronization.
 - Add another device.
 - Configure external AI.
- ✓ User has experienced value and understands the basic trust model.

Recovery

- Restart onboarding explanation from Settings.
- Continue local-only after denying permissions.
- Use sample data when no source permission is granted.

Device adaptation

Smartphone prioritizes receipt capture. Tablet offers capture, import, and sample workspace equally. Desktop prioritizes file import and device setup.

5. F02 — Capture a Receipt on Smartphone

Goal

Save a receipt quickly with little or no typing.

Entry points

- Global Capture action.
- Home-screen or lock-screen shortcut.
- Capture from Home.
- Capture from Records.

Flow

[Capture Options]

- (Choose Camera)

→ <Camera permission available?>
 No → [Permission Explanation] → (Allow or Choose Photo/Manual Entry)
 Yes → [Camera View]
 → {Detect receipt boundary, glare, blur, and missing corners}
 → <Image quality acceptable?>
 No → Show focused guidance → (Retake)
 Yes → (Take Photo)
 → [Image Confirmation]
 → (Use Photo or Retake)
 → {Encrypt and save original image locally}
 → {Run local extraction when available}
 → <External processing required or selected?>
 Yes → <Prior consent exists?>
 No → [External Processing Consent]
 → (Allow Once, Always Allow for This Feature, or Use Manual/Local Alternative)
 Yes → Continue.
 → [Processing]
 → {Extract merchant, date, amount, tax, payment source, and category}
 → {Check transaction match and duplicate probability}
 → <High confidence and no meaningful exception?>
 Yes → {Save and organize record}
 → [Saved Confirmation + Undo]
 ✓ Receipt is searchable and available for synchronization.
 No → <Can question be deferred?>
 Yes → {Save record and add exception to Review}
 → [Saved — Review Later]
 No → [One Focused Question]
 → (Select or enter answer)
 → {Complete record}
 → [Saved Confirmation + Undo]

Recovery

- Preserve original image if extraction fails.
- Retake image.
- Enter the amount and date manually.
- Save incomplete and review later.
- Undo save or restore a prior version.

6. F03 — Import a Digital Receipt through Share Sheet

Goal

Send a receipt, image, PDF, or email attachment from another application into Butlerly.

Flow

[External Application]

- (Share)
- (Select Butlerly)
- [Butlerly Import Preview]
- {Identify item type and source application}
- <Supported format?>
 - No → [Unsupported Format]
 - Offer Save Original, Open as Document, or Cancel.
 - Yes → Show file name, preview, processing location, and destination.
- (Import)
- {Encrypt and save original locally}
- {Extract and classify}
- <Possible matching transaction or existing receipt?>
 - Yes → [Link or Duplicate Review]
 - No → Continue.
- <High confidence?>
 - Yes → [Imported Confirmation + Undo]
 - No → Save and add to Inbox or ask one blocking question.
- ✓ User returns to the source application or opens the imported Butlerly record.

7. F04 — Import Files or Statements

Goal

Import one or many financial files without manually reorganizing them.

Primary devices

Desktop first for bulk import; tablet for moderate import; smartphone for small files.

Flow

[Import Center]

- (Choose Files or Drag and Drop)
- {Identify formats, file count, approximate item count, and duplicates}
- [Import Preview]
 - Files accepted.
 - Files unsupported.
 - Expected processing method.
 - Local versus external processing.
- (Start Import)
- {Create encrypted import batch and preserve source files}
- [Progress]
- <User leaves screen?>
 - Yes → Continue in background where supported; show status in Home/Activity.

- {Parse, normalize, match, classify, and detect duplicates}
- [Import Summary]
 - Imported.
 - Linked.
 - Skipped duplicates.
 - Needs review.
 - Failed.
- <Failures present?>
 - Yes → (Review Failures, Retry, or Export Report)
- <Exceptions present?>
 - Yes → (Review Now or Later)
- [Completed]
- ✓ Valid records are searchable and queued for synchronization.

Recovery

- Undo an entire import batch when no later dependencies prevent it.
- Retry failed files individually.
- Change mapping for structured files.
- Retain original source files.
- Export an import report.

8. F05 — AI Processing and Automatic Completion

Goal

Allow Butlerly to complete routine interpretation without requiring repetitive confirmation.

Flow

[New or Changed Record]

- {Gather approved local context}
- {Extract or infer candidate values}
- {Calculate confidence and consequence}
- <Action reversible and confidence high?>
 - Yes → Apply automatically.
 - Record reason, evidence, model/provider, timestamp, and prior value.
 - Offer Undo when surfaced to user.
 - No → <Confidence medium and consequence low?>
 - Yes → Save proposal and place in review queue.
 - No → Ask one focused question or require explicit confirmation.
- <External AI needed?>
 - Yes → Verify feature-specific consent before transfer.
- {Learn from correction if enabled}
- ✓ Record is completed, proposed, or routed to review with an understandable reason.

Required explanation format

- What Butlerly believes.
- Why it believes it.
- Confidence: High, Medium, or Low.
- What action was taken or recommended.
- How to correct or undo.

9. F06 — Review a Low-Confidence Item

Goal

Resolve uncertainty with the smallest reasonable amount of user effort.

Flow

[Review]

→ (Open Needs Review)

→ [Review Item]

- Plain-language issue.
- Butlerly recommendation.
- Evidence.
- Confidence.
- One primary decision.

→ (Confirm, Correct, Defer, or Inspect Source)

→ <Confirm?>

Yes → {Apply proposal and record decision}

→ <Correct?>

Yes → [Focused Edit]

→ (Save)

→ {Apply correction and optionally learn}

→ <Defer?>

Yes → (Choose optional reminder or reason)

→ Move to Deferred.

→ [Next Item or Completion]

✓ Item is resolved or intentionally deferred.

Tablet and desktop

Use list-detail layout. Desktop supports keyboard navigation and safe batch actions for genuinely similar items.

10. F07 — Resolve a Possible Duplicate

Goal

Prevent duplicate records while preserving evidence and user control.

Flow

[Possible Duplicate]

- Show both records side by side or in a comparison view.
- Highlight matching and differing fields.
- Butlerly recommends:
 - Merge.
 - Keep both.
 - Link without merging.
 - Replace lower-quality source.
- (Choose action)
- <Merge selected?>
 - Yes → {Combine relationships, source files, notes, and history}
 - Preserve both original source artifacts.
 - Create one canonical record.
 - [Merged + Undo]
- <Keep both?>
 - Yes → Mark as reviewed and not duplicate.
- <Link?>
 - Yes → Preserve separate records with explicit relationship.
- ✓ Duplicate risk is resolved without silent deletion.

11. F08 — Search and Retrieve a Record

Goal

Find a receipt, transaction, document, or related answer within seconds.

Flow

[Global Search]

- (Enter natural language, voice, or filters)
- {Parse merchant, dates, amounts, categories, locations, and record types}
- {Search local index first}
- [Results]
 - Best matches.
 - Match reasons.
 - Relevant preview.
 - Attention status when applicable.
- <Correct result visible?>
 - Yes → (Open)
 - [Record Detail]
 - User may inspect source, related records, explanation, or history.
 - No → (Refine, broaden, reset filters, or browse Records)
- <No results?>

- Yes → Explain searched scope and unavailable sources; offer alternatives.
- ✓ User finds the intended record or understands why it is unavailable.

State preservation

Returning from Record Detail restores the same query, filters, scroll position, and selected result.

12. F09 — Ask Butlerly About a Record

Goal

Understand a record without interpreting raw fields.

Flow

[Record Detail]

- (Ask Butlerly)
- [Contextual Assistant]
- (Enter question or select suggestion)
- {Use the selected record and approved related context}
- <External AI required?>
 - Yes → Confirm consent or use local alternative.
- {Generate answer with evidence and uncertainty}
- [Answer]
 - Direct conclusion.
 - Supporting evidence.
 - Confidence or missing information.
 - Suggested next action.
- (Inspect Source, Correct Record, Ask Follow-up, or Close)
- ✓ User understands the record or the remaining uncertainty.

The Assistant must distinguish observation, inference, and recommendation.

13. F10 — Edit, Undo, and Restore

Goal

Correct mistakes without losing prior information.

Immediate undo flow

(User performs reversible action)

- {Apply action and store prior state}
- [Undo Banner]
- <Undo selected before expiration?>
 - Yes → Restore prior state and synchronize reversal.

No → Prior state remains in version history according to retention policy.

Edit flow

[Record Detail]

- (Edit)
- [Focused Edit]
- Modify one or more fields.
- (Save)
- {Create version with device ID and timestamp}
- {Queue synchronization}
- [Saved]

Restore flow

[Version History]

- (Select prior version)
- [Compare Current and Prior]
- (Restore Selected Fields or Entire Version)
- {Create a new current version; do not erase history}
- [Restored]
- ✓ Correct value becomes active and remains recoverable.

14. F11 — Add a Trusted Device

Goal

Use Butlerly on another device while preserving encrypted local-first ownership.

Entry

Settings → My Devices → Add Device.

Flow A — Apple synchronization

[Add Device]

- (Choose Apple Sync)
- Butlerly explains what will synchronize and that each device keeps a local encrypted copy.
- (Enable)
- {Authenticate with platform-supported identity and device security}
- {Create or retrieve device-specific encryption keys}
- {Transfer encrypted data and change history}
- [Sync Progress]
- [Device Ready]
- ✓ New device appears in Trusted Devices.

Flow B — QR pairing

[Existing Trusted Device: Add Device]

- Display short-lived QR code.
- [New Device: Scan Code]
- {Mutual device authentication}
- Existing device shows new-device name and fingerprint.
- (Approve or Reject)
- {Exchange encrypted keys and approved data}
- [Device Ready]

Flow C — AirDrop/manual transfer

[Export Encrypted Device Package]

- (Send through AirDrop or Files)
- [New Device: Open Package]
- Authenticate and enter package secret if required.
- {Import identity and encrypted data}
- [Device Ready]

15. F12 — Synchronize and Merge Changes

Goal

Keep trusted devices consistent without requiring the user to manage synchronization.

Normal flow

{Local record changes}

- Create change set containing record ID, changed fields, values, device ID, and timestamps.
- Encrypt change set.
- Send through approved synchronization transport.
- Receiving device authenticates source and decrypts.
- <Different record?>
 - Yes → Apply independently.
- <Same record, different fields?>
 - Yes → Merge fields automatically.
- <Same record, same field?>
 - Yes → Compare valid timestamps.
 - Apply the most recent value.
 - Preserve superseded value in version history when meaningful.
- Update local index and status.
- ✓ All devices converge without unnecessary user intervention.

Exceptional flow

<Automatic merge cannot preserve meaning or integrity?>

→ Create Inbox item with both versions, source devices, timestamps, and recommendation.

→ User chooses value or manual merge.

Deletion flow

{Record deleted}

→ Create synchronized deletion marker.

→ Offline device later reconnects.

→ Deletion marker prevents old record from being restored as new.

→ Record remains in Recently Deleted according to retention policy.

16. F13 — Revoke a Lost Device

Goal

Remove access for one device without disabling the others.

Flow

[My Devices]

→ (Select Device)

→ [Device Detail]

→ (Revoke Access)

→ Butlerly explains consequences.

→ (Authenticate and Confirm)

→ {Mark device key as revoked}

→ {Stop accepting new changes from that key}

→ {Notify remaining trusted devices}

→ [Revoked]

→ Offer optional guidance for changing external provider credentials when relevant.

✓ Remaining devices continue normally.

17. F14 — Use Butlerly Offline

Goal

Continue core work without connectivity.

Flow

[Application Launch]

→ {Open encrypted local database}

→ <Network unavailable?>

Yes → Show quiet offline status.

- User can:
 - Capture receipts.
 - Search local data.
 - Edit records.
 - Review locally available exceptions.
 - Use local AI when available.
- {Store changes in local queue}
- <Connected-only feature requested?>
 - Yes → Explain limitation and offer local or later alternative.
- Connectivity returns.
- {Resume synchronization automatically}
- Apply merge rules from F12.
- [Changes Available on Devices]
- ✓ No local work is lost or blocked.

18. F15 — Manage Privacy and External AI Consent

Goal

Understand and control where data is stored and processed.

Contextual consent flow

(User invokes feature requiring external processing)

- [Consent Sheet]
 - What data will be sent.
 - Which provider receives it.
 - Why it is needed.
 - Whether the result is stored.
 - Available local/manual alternative.
- (Allow Once, Allow for This Feature, Not Now)
- Record the decision.
- Continue or use alternative.

Privacy settings flow

[Settings → Privacy & Data]

- View:
 - Local data.
 - Synchronized data.
 - Connected sources.
 - External AI providers.
 - Trusted devices.
- (Open category)
- Inspect, disable, export, delete, or change consent.

- ✓ User can understand and modify every material data boundary.

19. F16 — Export Data

Goal

Obtain a portable copy of Butlerly data.

Flow

[Privacy & Data → Export]

→ Choose scope:

- All data.
- Selected date range.
- Selected record types.
- Selected records.

→ Choose format:

- Butlerly encrypted archive.
- Human-readable files.
- Structured machine-readable export.

→ Choose whether to include originals and version history.

→ (Create Export)

→ {Authenticate for sensitive export}

→ {Generate locally when practical}

→ [Export Ready]

→ (Save to Files, AirDrop, or approved destination)

- ✓ User receives an understandable, portable export.

20. F17 — Delete and Restore Data

Delete record flow

[Record Detail]

→ (Delete)

→ Explain whether related records remain and how long recovery is available.

→ (Confirm)

→ {Move to Recently Deleted and create deletion marker}

→ [Deleted + Undo]

Restore flow

[Recently Deleted]

→ (Select Record)

→ (Restore)

→ {Restore canonical identity and relationships without duplication}

- {Synchronize restoration as a new change}
- [Restored]

Erase all data flow

[Privacy & Data → Erase All Data]

- Explain local, synchronized, connected-source, backup, and device consequences.
- Require device authentication and explicit confirmation.
- {Revoke synchronization keys as required}
- {Delete data according to documented retention behavior}
- [Erasure Confirmation]

21. F18 — Recover from Failure

21.1 Receipt extraction failure

[Processing Failed]

- State that the image is preserved.
- State whether data left the device.
- Offer Retake, Manual Entry, Try Another Method, or Review Later.

21.2 Import failure

[Import Summary]

- Show failed file and reason.
- Offer Retry, Change Mapping, Keep Original Only, or Export Report.

21.3 Synchronization failure

[Sync Delayed]

- Local editing remains enabled.
- Explain which devices may be out of date.
- Offer Retry, Inspect Pending Changes, Apple Sync Settings, or Manual Transfer.

21.4 External AI unavailable

[Provider Unavailable]

- Preserve task and input.
- Offer Local Processing, Manual Completion, Another Approved Provider, or Try Later.

21.5 Storage full

[Storage Required]

- Explain what is already preserved.

→ Offer Manage Local Storage, Export Archive, Remove Cached Copies, or Cancel.

Every error must communicate:

- What failed.
- What was preserved.
- Whether data left the device.
- What the user can do next.
- Whether retrying is safe.

22. Device-Specific Interaction Requirements

22.1 Smartphone

- One-handed primary actions where practical.
- Single-column focused flows.
- Camera and share-sheet integration.
- One exception at a time by default.
- Bottom navigation hidden only during immersive capture or authentication.
- Voice input available for search and assistant where supported.

22.2 Tablet

- List-detail split views.
- Side-by-side receipt and transaction comparison.
- Touch, keyboard, and pointer support.
- Drag and drop for imports and linking.
- Preserve selected item while navigating lists.

22.3 Desktop

- Persistent sidebar and global toolbar.
- Keyboard-first navigation and shortcuts.
- Multi-select and safe batch actions.
- Drag and drop and large-file import.
- Multi-pane review, search, and settings.
- More visible export, diagnostics, and advanced filters.

23. Accessibility Requirements for Every Flow

- Focus order follows the visual and logical sequence.
- Every action has an accessible name.
- Status changes are announced without stealing focus unnecessarily.
- Color is never the only indication of confidence, error, or completion.
- Touch targets meet platform accessibility guidance.

- Motion can be reduced.
- Dialogs return focus to the initiating control.
- Keyboard users can complete every desktop and tablet flow.
- Screen-reader users receive concise descriptions of receipt images, comparisons, and changed fields.
- Time-limited actions such as Undo provide an alternative through history when expiration would create loss.

24. Analytics and UX Measurement Events

Privacy-preserving product analytics may record flow-level events only when allowed by the user and company policy.

Recommended events:

- Flow started.
- Permission explanation viewed.
- Permission granted or deferred.
- Capture completed or abandoned.
- Import completed, partially completed, or failed.
- Exception confirmed, corrected, or deferred.
- Search produced result, no result, or reformulation.
- Undo used.
- Restore completed.
- Device pairing completed or failed.
- Sync merge completed or escalated.
- Export completed.

Analytics must not capture receipt contents, financial values, search text, or private document data unless explicitly necessary, consented to, and governed by a separate approved standard.

25. Dependencies and Governance

UX-0004 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- Product MVP scope and personas.
- Architecture decisions for local storage, encryption, identity, synchronization, and AI abstraction.

UX-0004 governs:

- UX-0005 — Screen and State Inventory.
- UX-0006 — AI Interaction and Explainability Specification.
- UX-0008 — Cross-Device Experience and Continuity.

- UX-0010 — Low-Fidelity Wireframe Specification.
- Prototype tasks and engineering acceptance tests.

26. Acceptance Criteria

This document is complete when:

- Every must-have Version 1 journey has an actionable flow.
- Each flow defines normal, decision, exception, recovery, and completion paths.
- Smartphone, tablet, and desktop adaptations are explicit.
- Local-first, offline, encryption, consent, device trust, and synchronization behavior are represented.
- Same-record merge and timestamp conflict rules are unambiguous.
- AI proposals, confidence, explanation, correction, and undo are represented.
- Designers can create screen states without inventing missing decisions.
- Engineers can derive functional acceptance tests from the flows.
- Usability tests can measure completion, error, recovery, and comprehension.

BDS-0000 / Finance V1 Scope Conformance

BDS-0000 — Butler Design System v1.0 governs visual tokens, concrete theme values, typography, spacing/radius, responsive breakpoints, component/state behavior, accessibility implementation, localization/original-data presentation, and Flutter UI constraints. Finance V1 visible primary navigation is Home, Transactions, Review, Search, and Settings; Assistant is optional/non-primary. Legacy flow wording or IDs may remain for traceability but must not change the visible destination model.

F11, F12, F13 and other trusted-device/synchronization flows are future/optional design references, not Finance V1 MVP requirements, unless product scope explicitly enables cross-device capability. Core local workflows must remain complete without an account, pairing, or synchronization. UI-authored text follows the selected application language; source/user/imported/scanned data preserves its original language by default, and original currency/amount remains authoritative with any reference conversion shown separately.